

DOCUMENTATION

An Internship Report on Android Application Development

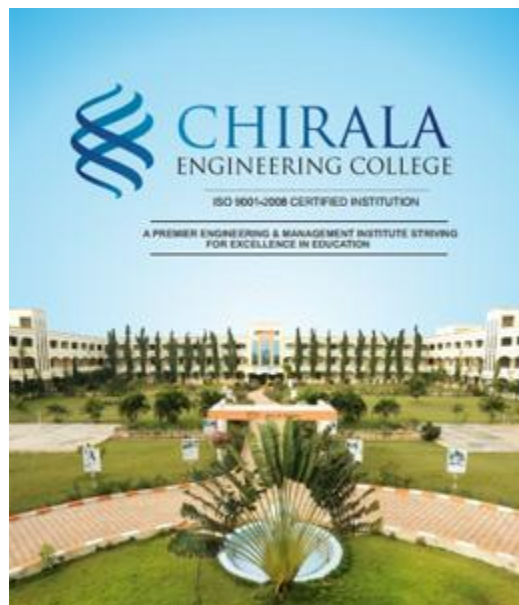
Submitted in partial fulfilment of the requirements for the award of Bachelor of Technology (B.TECH) in “COMPUTER SCIENCE & ENGINEERING (IT)”

SUBMITTED BY:

K .Tagore Reddy --- 22E91A0553 (CSE)

G .Ajay Krishna --- 22E91A1209 (IT)

P .Susmitha Padma --- 22E91A1220 (IT)



❖ OFFICIAL CERTIFICATION:

This is to certify that “Tagore Reddy, Ajay Krishna, Susmitha Padma” Regd.No. 22E91A0553, 22E91A1209, 22E91A1220 has completed his/her Internship in Andhra Pradesh State Skill Development Corporation (APSSDC) on “Android Application Development” under my supervision as a part of partial fulfilment of the requirement for the Degree of III B.Tech in the Department of CSE & IT of Chirala Engineering College . This is accepted for evaluation.

Head of the Department

External Examiner

❖ CERTIFICATE FROM INTERNSHIP ORGANIZATION



❖ Student Declaration:

I, AJAY KRISHNA & TAGORE REDDY a student of “Android Application Development Program”, Reg. No. 22E91A1220, 22E91A1209& 22E91A0553 of the Department of Computer Science & Engineering and Information Technology hereby declare that I have completed the mandatory internship from 03-06-2024 to 27-07-2024 in Andhra Pradesh State Skill Development Corporation under the Department of Computer Science& Engineering in Chirala Engineering College.

(Signature of the student and Date)

❖ ACKNOWLEDGEMENT:

We wish to express our sincere gratitude to **Dr. Donepudi Jagadish, Principal**, Chirala Engineering College, Chirala, for his constant support and encouragement to complete the project.

We extremely grateful to **Head of the Computer Science & Engineering**, Chirala Engineering College, Chirala, for kind encouragement and support throughout our project.

We thank **all teaching and non-teaching staff** of the department of C.S.E, Chirala Engineering College, Chirala, for their constant support and co-operation.

Finally, we express our sincere thanks to our **parents, family members and friends** who gave us moral support in completing our project successfully.

Submitted by,

K .Tagore Reddy --- 22E91A0553 (CSE)

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❖ OVERVIEW OF THE ORGANIZATION

Vision and Mission:

Vision of the institute	To be a world-class leader in transforming lives through an innovative, rigorous and compassionate approach by imparting high-quality academic excellence in technical education to uplift the living standards of the rural youth by promoting cutting edge technologies, employability, higher education, and research with socio-technical, eco-friendly, and entrepreneurial values.
Mission of the institute	To develop high quality technical personal with a sound footing on basic engineering principles, innovative capabilities and exemplary professional conduct to lead and to use technology for the progress of mankind, training and adopting them to changing technological environment by proving the high quality instruction, infra, faculty, effective training and learning methods imbibing socio-ethical, eco-friendly and entrepreneurial values as the inner strength for the rural development.
Vision of the Department	To become a center for academic excellence by producing high quality engineers in Computer Science& Engineering with global competence empowered by technical expertise, innovation, research and the knowledge of cutting edge technologies, employability, higher education, entrepreneurial skills along with socio-ethical and eco-friendly values for the rural development.
Mission of the Department	To develop the high quality Computer Science & Engineering with the fundamental engineering principles, intellectually adept, professionally deft, innovative research capabilities, exemplary skills, technological progress of mankind, tracking technological environment by providing the high quality instruction, infra, faculty, modern teaching and learning methods, training with socio-ethical and entrepreneurial values as the inner strength for the rural development.

Programme Outcomes (PO's):

Engineering Graduates will be able to:

1. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to

comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

Programme Specific Outcome (PSO's):

PSO1: Should have the capability to comprehend the technological advancements in the usage of modern design tools to analyze and design subsystems/processes for a variety of applications in Electronics & Communication Engineering.

PSO 2: An understanding of social-awareness & environmental-wisdom along with successful career and to sustain passion and zeal for real-world applications using optimal resources as an Entrepreneur.

Program Educational Objectives (PEOs):

PEO1: Preparation: To strengthen the learners with fundamental knowledge in the concepts of Engineering Mathematics, Applied Science and Basic Electronics Engineering subjects for problem analysis, comprehend to develop solutions for new hardware, software, and other tools in Electronics & Communication Engineering field.

PEO2: Core Competence: To enable the learners with core curriculum knowledge in theory and practical of Electronics & Communication Engineering to develop the innovative skills, critical analysis, design, simulation, investigation of complex problems, reasoning, development & testing knowledge for offering solutions to real time problems related to globally evolving techno-corridor developments of Electronics & Communication Engineering.

PEO3: Breadth Knowledge: To enable the learners with breadth knowledge to build the Electronics & Communication Engineering professionals to have the team work and skills for developing communicative abilities, lifelong learning and

aptitude of project management & finance with entrepreneurial values for rural development.

PEO4: Advanced Professional Knowledge: Implementation of the multi-disciplinary approach, development of the R&D skills by MOUs with premier industries and institutions, interacting with training sessions and industrial visits to the learners to have awareness in latest trends of Communication, Networks, Signal processing etc. with the modern tools and modules compatible to escalating needs of the society.

PEO5: Career Development and Ethics: To enable the learners with the aptitude of competitive knowledge of real-time requirement of cutting edge technologies for promoting employability, higher education by imbibing ethical, social and eco-friendly values.

ABSTRACT

The BOY FLYING GAME is a “2D Animation Game Design”. A young adventurer as he embarks on a magical journey, taking flight through landscapes filled with wonder and excitement.

INTRODUCTION:

Discover a vibrant, imaginative world where the laws of gravity are defied, and the possibilities are endless. Source effortlessly by guiding the boy with simple, responsive controls that allow you to maneuver through the skies with grace.

Feast your eyes on breathtaking environments, where each scene is a masterpiece of color, light, and attention to detail.

KEY FEATURES:

- **Game Exploration:** Freely explore a Challenge Obstacle vast, detailed world filled with diverse environments.
- **Challenge Obstacles:** The flight boy tires to kill all birds appear before him & it doesn't have any time limit.
- **Overview:** Discover the captivating story behind the boy's journey, as you navigate through a world of wonder and imagination.

USER INTERFACE:

The user interface is designed to be seamless and unobtrusive, allowing you to focus on the gameplay. On-screen prompts and tutorials ensure a smooth, guided experience for both new and experienced players.

USAGE OF 2D ANIMATION:

2D animation combines graphic design, motion graphics, voiceovers, and at times character animation to create visually appealing videos that tell a story and visualize complex topics.

RULES OF THE GAME:

- ✓ If you touch right side of the screen then the flight will shoot the bullets.
- ✓ If you touch left side of the screen then we can control the flight to move up and down.
- ✓ If the boy misses any bird to kill (Or) bird passes through the boy side then the game will be over.
- ✓ If the flight boy and flying bird collide (Or) hit each other then also game will be over.

ANDROID STUDIO

➤ INTRODUCTION:

Android studio is the official integrated development

Environment (IDE) for Google's android operating system, built on Jet Brains' IntelliJ IDEA software and designed specifically for Android development. It was

announced in 2013 at the Google I/O conference and has since become the primary tool for Android app development.

➤ **KEY FEATURES OF ANDROID STUDIO:**

1) INTELLIGENT CODE EDITOR:

Offers advanced code completion, refactoring, and real time code analysis.

2) FAST EMULATOR:

Provides a fast and feature-rich emulator to test and debug apps in various environments.

3) FLEXIBLE BUILD SYSTEM:

Based on Gradle, it allows for more custom build configuration and automation.

4) LAYOUT EDITOR:

A powerful tool for designing and previewing app UIs, with support for drag-and-drop components and real-time rendering.

➤ **TOOLS USED IN ANDROID STUDIO**

1) ANDROID EMULATOR:

Allows you to test applications on a virtual device.

2) LOGCAT:

A logging system that provides a way to view log measures from your app.

3) ANDROID SDK:

A collection of tools and libraries for Android development, including platform tools, build tools, and system images.

4) GRADLE:

A build automation tool used to compile, build, and package your app.

5) LAYOUT EDITOR:

A visual editor to design your app's user interface.

6) Profiler:

Tools for profiling the performance of your app, including CPU, memory, network, and energy usage.

7) LINT:

A static code analysis tool that helps identify and correct problems with the structural quality of your code.

8) FIRE BASE:

Tools for integrating Firebase services like analytics, authentication, and cloud messaging.

9) CONSTRAINT LAYOUT:

A flexible layout manager used to build complex layouts without nested view hierarchies.

10) VECTOR ASSET STUDIO:

Helps to import vector graphics into your project.

11) APK ANALYSER:

Helps analyze the contents of your APK to reduce its size.

12) AVD MANAGER:

Manages android virtual devices (AVDs) that can run the Android Emulator.

These tools and features collectively make Android studio a powerful integrated development environment (IDE) for Android development.

➤ ANDROID VERSIONS IN ALPHABETICAL ORDER:

A-Alpha

B-Beta

C-Cupcake

D-Donut

E-Éclair

F-Froye

G-GingerBread

H-HoneyComb

I-Ice Cream Sandwich

J-Jellybean

K-Kit Kat

L-Lollypop

M-Marshmallow

N-Nought

O-Oreo

P-pie

Android 10, 11,12,13,14

➤ PROGRAMMING LANGUAGES USED:

1) JAVA:

The traditional and original languages for android development.

2) KOTLIN:

Officially supported by Google since 2017, and now the preferred language for Android development.

3) C++:

Used with the Android NDK (Native Development kit) for high-performance applications that require native code.

You can also use other languages through various plugins and frameworks, such as Dart with Flutter, but java and Kotlin remain the primary languages.

➤ CONCLUSION:

Android studio is the official integrated development environment (IDE) for Android application development, offering a comprehensive set of tools and features to streamline the development process. It supports a wide range of coding, debugging, and performance tooling to a flexible build system. With its strong integration with the Android SDK, a rich layout editor, and extensive support for Kotlin, java, and C++, Android studio is a powerful and versatile platform for building high-quality Android app. whether you're a beginner or an experienced developer.

SAMPLE PROJECT

TITLE: THE BOY FLYING GAME APP



DESCRIPTION

The BOY FLYING GAME is a “2D Animation Game Design”. A young adventurer as he embarks on a magical journey, taking flight through landscapes filled with wonder and excitement. Discover a vibrant, imaginative world where the laws of gravity are defied, and the possibilities are endless. Source effortlessly by guiding

the boy with simple, responsive controls that allow you to maneuver through the skies with grace. Feast your eyes on breathtaking environments, where each scene is a masterpiece of color, light, and attention to detail.

SPLASH SCREEN:

A splash screen is a graphical introduction that appears when you launch an application or software. Splash screens are used to create a first impression and provide feedback to the user that the application is loading. It typically displays the app's logo, name.

➤ activity_splash.xml :(CODE)

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/splash"
    tools:context=".Splash">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="300dp"
        android:fontFamily="@font/eagle_lake"
```

```
android:gravity="center"  
android:text="Welcome to \n Game"  
android:textColor="@color/white"  
android:textColorHint="@color/white"  
android:textSize="40sp" />
```

```
</RelativeLayout>
```

DESIGN:



➤ activity_activity.xml: (CODE)

```
<?xml version="1.0" encoding="utf-8"?>  
<RelativeLayout  
  xmlns:android="http://schemas.android.com/apk/res/android"  
  xmlns:app="http://schemas.android.com/apk/res-auto"  
  xmlns:tools="http://schemas.android.com/tools"  
  android:layout_width="match_parent"
```

```
android:layout_height="match_parent"  
tools:context=".MainActivity">
```

```
<ImageView
```

```
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:src="@drawable/background"  
    android:scaleType="centerCrop"/>
```

```
<TextView
```

```
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:textSize="48sp"  
    android:text="Boy Flying Game"  
    android:textColor="@color/black"  
    android:layout_above="@id/play"  
    android:layout_below="@id/highScoreTxt"  
    android:layout_centerHorizontal="true"/>
```

```
<Button
```

```
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Play"  
    android:backgroundTint="#6BCCF8"  
    android:textSize="50sp"  
    android:layout_centerInParent="true"  
    android:textColor="@color/black"  
    android:id="@+id/play"/>
```

<TextView

```
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:textSize="24sp"  
    android:textColor="@color/black"  
    android:text="High Score"  
    android:id="@+id/highScoreTxt"  
    android:layout_margin="16dp"  
    android:layout_alignParentRight="true"/>
```

<ImageView

```
    android:layout_width="36dp"  
    android:layout_height="36dp"  
    android:layout_alignParentBottom="true"  
    android:layout_margin="16dp"  
    android:id="@+id/volumeCtrl"  
    android:src="@drawable/ic_volume_up_black_24dp"/>
```

</RelativeLayout>

DESIGN:



➤ MainActivity.java:(CODE)

```
package com.example.boyflyinggame;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.content.Intent;
```

```
import android.content.SharedPreferences;
```

```
import android.os.Bundle;
```

```
import android.view.View;
```

```
import android.view.WindowManager;
```

```
import android.widget.ImageView;
```

```
import android.widget.TextView;
```

```
public class MainActivity extends AppCompatActivity {

    private boolean isMute;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        getWindow().setFlags(WindowManager.LayoutParams.FLAG_FULL
SCREEN,
        WindowManager.LayoutParams.FLAG_FULLSCREEN);

        setContentView(R.layout.activity_main);

        findViewById(R.id.play).setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new Intent(MainActivity.this,
GameActivity.class));
            }
        });

        TextView highScoreTxt = findViewById(R.id.highScoreTxt);

        final SharedPreferences prefs =
getSharedPreferences("game", MODE_PRIVATE);
        highScoreTxt.setText("HighScore: " +
prefs.getInt("highscore", 0));
    }
}
```

```
isMute = prefs.getBoolean("isMute", false);

final ImageView volumeCtrl = findViewById(R.id.volumeCtrl);

if (isMute)

volumeCtrl.setImageResource(R.drawable.ic_volume_off_black_24
dp);
else

volumeCtrl.setImageResource(R.drawable.ic_volume_up_black_24
dp);

volumeCtrl.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {

        isMute = !isMute;
        if (isMute)

volumeCtrl.setImageResource(R.drawable.ic_volume_off_black_24
dp);
        else

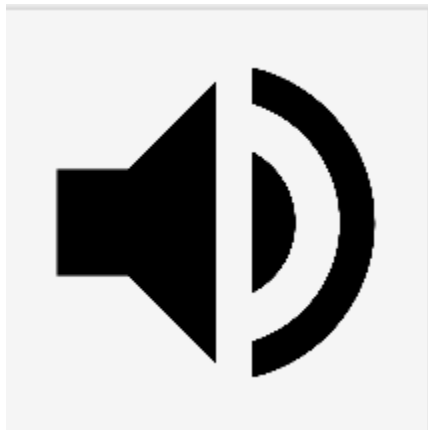
volumeCtrl.setImageResource(R.drawable.ic_volume_up_black_24
dp);

        SharedPreferences.Editor editor = prefs.edit();
        editor.putBoolean("isMute", isMute);
```

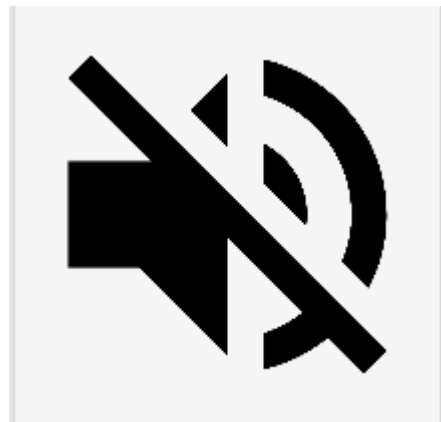
```
        editor.apply();  
    }  
});  
  
}  
}
```

In this main activity we contain volume control code.

UNMUTE:



MUTE:



➤ **Splash.java:(CODE)**

```
package com.example.boyflyinggame;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```

import android.content.Intent;
import android.os.Bundle;
import android.os.Handler;

public class Splash extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_splash);
        new Handler().postDelayed(new Runnable() {
            @Override
            public void run() {
                Intent intent=new Intent(Splash.this,
MainActivity.class);
                startActivity(intent);
                finish();
            }
        },4000);
    }
}

```

➤ GameView.java :(CODE)

```

package com.example.boyflyinggame;

import android.content.Context;

```

```
import android.content.Intent;
import android.content.SharedPreferences;
import android.graphics.Canvas;
import android.graphics.Color;
import android.graphics.Paint;
import android.graphics.Rect;
import android.media.AudioAttributes;
import android.media.AudioManager;
import android.media.SoundPool;
import android.os.Build;
import android.view.MotionEvent;
import android.view.SurfaceView;
```

```
import java.util.ArrayList;
import java.util.List;
import java.util.Random;
```

```
public class GameView extends SurfaceView implements
Runnable {
```

```
    private Thread thread;
    private boolean isPlaying, isGameOver = false;
    private int screenX, screenY, score = 0;
    public static float screenRatioX, screenRatioY;
    private Paint paint;
    private Bird[] birds;
    private SharedPreferences prefs;
    private Random random;
    private SoundPool soundPool;
    private List<Bullet> bullets;
```

```
private int sound;
private Flight flight;
private GameActivity activity;
private Background background1, background2;

public GameView(GameActivity activity, int screenX, int
screenY) {
    super(activity);

    this.activity = activity;

    prefs = activity.getSharedPreferences("game",
Context.MODE_PRIVATE);

    if (Build.VERSION.SDK_INT >=
Build.VERSION_CODES.M) {

        AudioAttributes audioAttributes = new
AudioAttributes.Builder()

.setContentType(AudioAttributes.CONTENT_TYPE_MUSIC)
.setUsage(AudioAttributes.USAGE_GAME)
.build();

        soundPool = new SoundPool.Builder()
.setAudioAttributes(audioAttributes)
.build();

    } else
```

```
    soundPool = new SoundPool(1,  
AudioManager.STREAM_MUSIC, 0);
```

```
    sound = soundPool.load(activity, R.raw.shoot1, 1);
```

```
    this.screenX = screenX;  
    this.screenY = screenY;  
    screenRatioX = 1920f / screenX;  
    screenRatioY = 1080f / screenY;
```

```
    background1 = new Background(screenX, screenY,  
getResources());
```

```
    background2 = new Background(screenX, screenY,  
getResources());
```

```
    flight = new Flight(this, screenY, getResources());
```

```
    bullets = new ArrayList<>();
```

```
    background2.x = screenX;
```

```
    paint = new Paint();  
    paint.setTextSize(128);  
    paint.setColor(Color.WHITE);
```

```
    birds = new Bird[4];
```

```
    for (int i = 0; i < 4; i++) {
```

```
        Bird bird = new Bird(getResources());
```

```

        birds[i] = bird;

    }

    random = new Random();

}

@Override
public void run() {

    while (isPlaying) {

        update ();
        draw ();
        sleep ();

    }

}

private void update () {

    background1.x -= 10 * screenRatioX;
    background2.x -= 10 * screenRatioX;

    if (background1.x + background1.background.getWidth() <
0) {
        background1.x = screenX;
    }
}

```

```
    if (background2.x + background2.background.getWidth() < 0) {  
        background2.x = screenX;  
    }
```

```
    if (flight.isGoingUp)  
        flight.y -= 30 * screenRatioY;  
    else  
        flight.y += 30 * screenRatioY;
```

```
    if (flight.y < 0)  
        flight.y = 0;
```

```
    if (flight.y >= screenY - flight.height)  
        flight.y = screenY - flight.height;
```

```
List<Bullet> trash = new ArrayList<>();
```

```
for (Bullet bullet : bullets) {
```

```
    if (bullet.x > screenX)  
        trash.add(bullet);
```

```
    bullet.x += 50 * screenRatioX;
```

```
for (Bird bird : birds) {
```

```
    if (Rect.intersects(bird.getCollisionShape(),  
        bullet.getCollisionShape())) {
```

```

        score++;
        bird.x = -500;
        bullet.x = screenX + 500;
        bird.wasShot = true;

    }

}

}

for (Bullet bullet : trash)
    bullets.remove(bullet);

for (Bird bird : birds) {

    bird.x -= bird.speed;

    if (bird.x + bird.width < 0) {

        if (!bird.wasShot) {
            isGameOver = true;
            return;
        }

        int bound = (int) (30 * screenRatioX);
        bird.speed = random.nextInt(bound);

        if (bird.speed < 10 * screenRatioX)

```

```

        bird.speed = (int) (10 * screenRatioX);

        bird.x = screenX;
        bird.y = random.nextInt(screenY - bird.height);

        bird.wasShot = false;
    }

    if (Rect.intersects(bird.getCollisionShape(),
flight.getCollisionShape())) {

        isGameOver = true;
        return;
    }

}

}

private void draw () {

    if (getHolder().getSurface().isValid()) {

        Canvas canvas = getHolder().lockCanvas();
        canvas.drawBitmap(background1.background,
background1.x, background1.y, paint);
        canvas.drawBitmap(background2.background,
background2.x, background2.y, paint);

        for (Bird bird : birds)

```

```

        canvas.drawBitmap(bird.getBird(), bird.x, bird.y,
paint);

        canvas.drawText(score + "", screenX / 2f, 164, paint);

        if (isGameOver) {
            isPlaying = false;
            canvas.drawBitmap(flight.getDead(), flight.x, flight.y,
paint);

            getHolder().unlockCanvasAndPost(canvas);
            saveIfHighScore();
            waitBeforeExiting ();
            return;
        }

        canvas.drawBitmap(flight.getFlight(), flight.x, flight.y,
paint);

        for (Bullet bullet : bullets)
            canvas.drawBitmap(bullet.bullet, bullet.x, bullet.y,
paint);

        getHolder().unlockCanvasAndPost(canvas);

    }

}

private void waitBeforeExiting() {

```

```

        try {
            Thread.sleep(3000);
            activity.startActivity(new Intent(activity,
MainActivity.class));
            activity.finish();
        } catch (InterruptedException e) {
            e.printStackTrace();
        }
    }

    private void saveIfHighScore() {

        if (prefs.getInt("highscore", 0) < score) {
            SharedPreferences.Editor editor = prefs.edit();
            editor.putInt("highscore", score);
            editor.apply();
        }

    }

    private void sleep () {
        try {
            Thread.sleep(17);
        } catch (InterruptedException e) {
            e.printStackTrace();
        }
    }

    public void resume () {

```

```

        isPlaying = true;
        thread = new Thread(this);
        thread.start();
    }

    public void pause () {

        try {
            isPlaying = false;
            thread.join();
        } catch (InterruptedException e) {
            e.printStackTrace();
        }

    }

    @Override
    public boolean onTouchEvent(MotionEvent event) {

        switch (event.getAction()) {
            case MotionEvent.ACTION_DOWN:
                if (event.getX() < screenX / 2) {
                    flight.isGoingUp = true;
                }
                break;
            case MotionEvent.ACTION_UP:
                flight.isGoingUp = false;
                if (event.getX() > screenX / 2)

```

```

        flight.toShoot++;
        break;
    }

    return true;
}

public void newBullet() {

    if (!prefs.getBoolean("isMute", false))
        soundPool.play(sound, 1, 1, 0, 0, 1);

    Bullet bullet = new Bullet(getResources());
    bullet.x = flight.x + flight.width;
    bullet.y = flight.y + (flight.height / 2);
    bullets.add(bullet);

}
}

```

➤ GameActivity.java:(CODE)

```
package com.example.boyflyinggame;
```

```
import androidx.appcompat.app.AppCompatActivity;
import android.graphics.Point;
```

```
import android.os.Bundle;
import android.view.WindowManager;

public class GameActivity extends AppCompatActivity {

    private GameView gameView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        getWindow().setFlags(WindowManager.LayoutParams.FLAG_FULLSCREEN,
            WindowManager.LayoutParams.FLAG_FULLSCREEN);

        Point point = new Point();
        getWindowManager().getDefaultDisplay().getSize(point);

        gameView = new GameView(this, point.x, point.y);

        setContentView(gameView);
    }

    @Override
    protected void onPause() {
        super.onPause();
        gameView.pause();
    }
}
```

```

    }

    @Override
    protected void onResume() {
        super.onResume();
        gameView.resume();
    }
}

```

➤ Flight.java:(CODE)

```
package com.example.boyflyinggame;
```

```
import static
com.example.boyflyinggame.GameView.screenRatioX;
import static
com.example.boyflyinggame.GameView.screenRatioY;
```

```
import android.content.res.Resources;
import android.graphics.Bitmap;
import android.graphics.BitmapFactory;
import android.graphics.Rect;
```

```
public class Flight {

    int toShoot = 0;
    boolean isGoingUp = false;
    int x, y, width, height, wingCounter = 0, shootCounter = 1;

```

```

    Bitmap flight1, flight2, shoot1, shoot2, shoot3, shoot4,
    shoot5, dead;
    private GameView gameView;

    Flight (GameView gameView, int screenY, Resources res) {

        this.gameView = gameView;

        flight1 = BitmapFactory.decodeResource(res,
        R.drawable.fly1);
        flight2 = BitmapFactory.decodeResource(res,
        R.drawable.fly2);

        width = flight1.getWidth();
        height = flight1.getHeight();

        width /= 4;
        height /= 4;

        width = (int) (width * screenRatioX);
        height = (int) (height * screenRatioY);

        flight1 = Bitmap.createScaledBitmap(flight1, width, height,
        false);
        flight2 = Bitmap.createScaledBitmap(flight2, width,
        height, false);

        shoot1 = BitmapFactory.decodeResource(res,
        R.drawable.shoot1);
        shoot2 = BitmapFactory.decodeResource(res,

```

```

R.drawable.shoot2);
    shoot3 = BitmapFactory.decodeResource(res,
R.drawable.shoot3);
    shoot4 = BitmapFactory.decodeResource(res,
R.drawable.shoot4);
    shoot5 = BitmapFactory.decodeResource(res,
R.drawable.shoot5);

    shoot1 = Bitmap.createScaledBitmap(shoot1, width, height,
false);
    shoot2 = Bitmap.createScaledBitmap(shoot2, width,
height, false);
    shoot3 = Bitmap.createScaledBitmap(shoot3, width,
height, false);
    shoot4 = Bitmap.createScaledBitmap(shoot4, width,
height, false);
    shoot5 = Bitmap.createScaledBitmap(shoot5, width,
height, false);

    dead = BitmapFactory.decodeResource(res,
R.drawable.dead);
    dead = Bitmap.createScaledBitmap(dead, width, height,
false);

    y = screenY / 2;
    x = (int) (64 * screenRatioX);

}

Bitmap getFlight () {

```

```
if (toShoot != 0) {  
  
    if (shootCounter == 1) {  
        shootCounter++;  
        return shoot1;  
    }  
  
    if (shootCounter == 2) {  
        shootCounter++;  
        return shoot2;  
    }  
  
    if (shootCounter == 3) {  
        shootCounter++;  
        return shoot3;  
    }  
  
    if (shootCounter == 4) {  
        shootCounter++;  
        return shoot4;  
    }  
  
    shootCounter = 1;  
    toShoot--;  
    gameView.newBullet();  
  
    return shoot5;  
}
```

```

    if (wingCounter == 0) {
        wingCounter++;
        return flight1;
    }
    wingCounter--;

    return flight2;
}

Rect getCollisionShape () {
    return new Rect(x, y, x + width, y + height);
}

Bitmap getDead () {
    return dead;
}

}

```

In this flight.java file contain different flights that are acts as in different situation. One is Staring flight & second flight is full speed flight.

FLIGHT---1:



FLIGHT---2:



In this flight.java also contain dead flight. When bird & flight hit (Or) collide each other then the flight boy died.

DEAD FLIGHT:



➤ Bullet.java:(CODE)

```
package com.example.boyflyinggame;
```

```
import static com.example.boyflyinggame.GameView.screenRatioX;
```

```
import static com.example.boyflyinggame.GameView.screenRatioY;
```

```
import android.content.res.Resources;
```

```
import android.graphics.Bitmap;
```

```
import android.graphics.BitmapFactory;
```

```
import android.graphics.Rect;
```

```
public class Bullet {
```

```
    int x, y, width, height;
```

```
    Bitmap bullet;
```

```

Bullet (Resources res) {

    bullet = BitmapFactory.decodeResource(res,
R.drawable.bullet);

    width = bullet.getWidth();
    height = bullet.getHeight();

    width /= 4;
    height /= 4;

    width = (int) (width * screenRatioX);
    height = (int) (height * screenRatioY);

    bullet = Bitmap.createScaledBitmap(bullet, width, height,
false);

}

Rect getCollisionShape () {
    return new Rect(x, y, x + width, y + height);
}

}

```

In this bullet.java contain bullet image that is comes from flight.

BULLET:



In this we also contain different shoot images to Make difference in shooting time then user can easily attract to the game.

SHOOT—1:



SHOOT—2:



SHOOT—3:



SHOOT—4:



SHOOT—5:



➤ Bird.java:(CODE)

```
package com.example.boyflyinggame;
```

```
import android.content.res.Resources;
```

```
import android.graphics.Bitmap;
```

```
import android.graphics.BitmapFactory;
```

```
import android.graphics.Rect;
```

```
import static com.example.boyflyinggame.GameView.screenRatioX;
```

```
import static com.example.boyflyinggame.GameView.screenRatioY;
```

```
public class Bird {
```

```
    public int speed = 20;
```

```
    public boolean wasShot = true;
```

```
    int x = 0, y, width, height, birdCounter = 1;
```

```
Bitmap bird1, bird2, bird3, bird4;
```

```
Bird (Resources res) {
```

```
    bird1 = BitmapFactory.decodeResource(res,  
R.drawable.bird1);
```

```
    bird2 = BitmapFactory.decodeResource(res,  
R.drawable.bird2);
```

```
    bird3 = BitmapFactory.decodeResource(res,  
R.drawable.bird3);
```

```
    bird4 = BitmapFactory.decodeResource(res,  
R.drawable.bird4);
```

```
    width = bird1.getWidth();  
    height = bird1.getHeight();
```

```
    width /= 6;  
    height /= 6;
```

```
    width = (int) (width * screenRatioX);  
    height = (int) (height * screenRatioY);
```

```
    bird1 = Bitmap.createScaledBitmap(bird1, width, height,  
false);
```

```
    bird2 = Bitmap.createScaledBitmap(bird2, width, height,  
false);
```

```
    bird3 = Bitmap.createScaledBitmap(bird3, width, height,  
false);
```

```
    bird4 = Bitmap.createScaledBitmap(bird4, width, height,  
false);
```

```
    y = -height;  
}
```

```
Bitmap getBird () {
```

```
    if (birdCounter == 1) {  
        birdCounter++;  
        return bird1;  
    }
```

```
    if (birdCounter == 2) {  
        birdCounter++;  
        return bird2;  
    }
```

```
    if (birdCounter == 3) {  
        birdCounter++;  
        return bird3;  
    }
```

```
    birdCounter = 1;
```

```
    return bird4;  
}
```

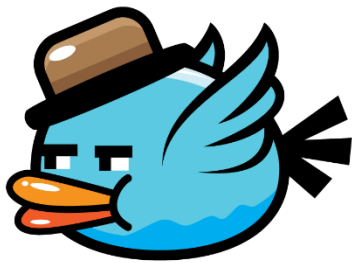
```
Rect getCollisionShape () {
```

```
    return new Rect(x, y, x + width, y + height);  
}
```

```
}
```

In this Bird.java code we contain 4- different types of birds to make difference in the game.

BIRD—1:



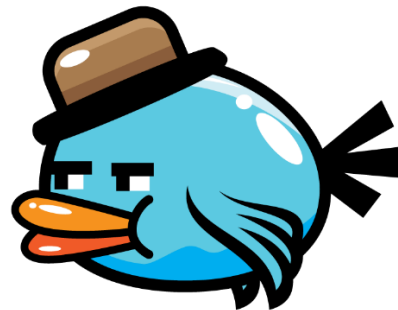
BIRD—1:



BIRD—3:



BIRD—4:



➤ Background.java:(CODE)

```
package com.example.boyflyinggame;
```

```
import android.content.res.Resources;
```

```
import android.graphics.Bitmap;
```

```
import android.graphics.BitmapFactory;
```

```
public class Background {
```

```
    int x = 0, y = 0;
```

```
    Bitmap background;
```

```
    Background (int screenX, int screenY, Resources res) {
```

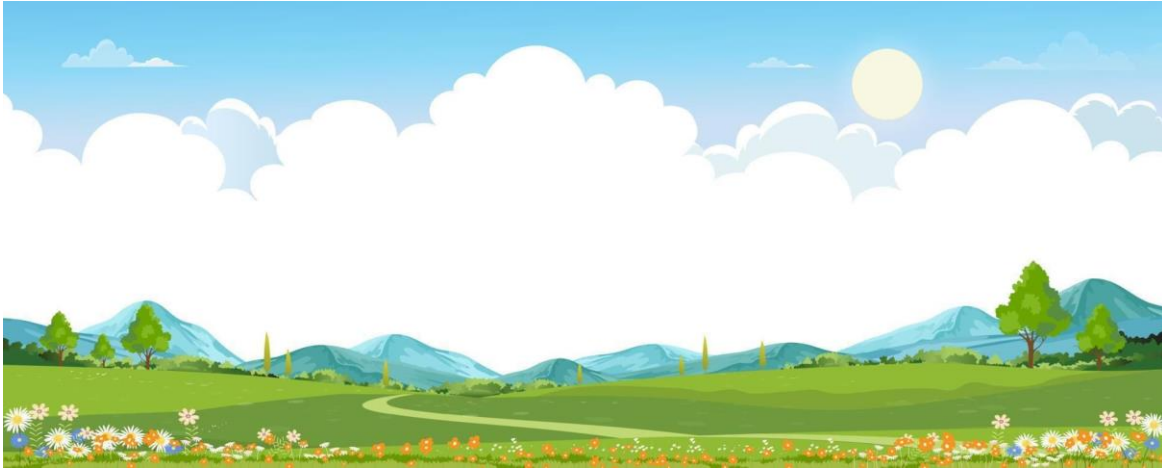
```
        background = BitmapFactory.decodeResource(res,  
R.drawable.background);
```

```
        background = Bitmap.createScaledBitmap(background,  
screenX, screenY, false);
```

```
    }
```

```
}
```

BACKGROUND IMAGE:



CONCLUSION:

The Boy Flying Game offers a truly impressive and captivating experience, blending stunning visuals, intuitive controls, and a sense of wonder and adventure. Embark on a magical journey with the young protagonist, soaring through a realm of imagination and discovery, where the only limit is your own aerial process.

And it is a Simple, Addictive, Challenging, & Endless game to attract the children. We hope you've enjoyed learning about the captivating world of the Boy Flying Game. Get ready to take to the skies and experience the thrill of defying gravity!

Thank
you